Q2: For $R_{XY} = \{(0,0),(0,1),(1,0),(1,1)\}$, if $f(0,0) = 0.1, f(0,1) = 0.2, f(1,0) = 0.4$ and $f(1,1) = 0.3$. Then $E(XY) =$

A) 1 B) 0.4 C) 0.8 (D) 0.3 E) None

Q3: For random variables X, Y . If $Corr(x,y) = 3$ and $\sigma_X \sigma_Y = 2$. Then $Cov(2x - 3,1 - 3y) =$

A) 24 B) -24 (C) -36 D) 36 E) none

Q4: Let X, Y be two independent random variables with $E(X) = E(Y) = 3$. Then $E(XY) =$

A) 4 (B) 9 C) 1 D) 16 E) None

Q5: If $P(A) = 0.6$ and $P(A \mid B) = P(B \mid A)$, then $P(B) =$

A) 0.1 B) 0.2 C) 0.4 D) 0.6 E) None

Question 6: Complete the following probability distribution to have two independent random variables. **(5 marks)**

X \ Y	0	1	$f_X(x)$
1	0.28	0.42	0.7
2	0.12	0.18	0.3
$f_Y(y)$	0.4	0.6	1

$f(x,y) = f_x(x) \cdot f_y(y)$

$f(1,1) = f_x(1) \cdot f_y(1)$ ..

Q2:

$$EXY = \sum X\,Y\,f(x,y)$$

$$= [0\,f(0,0) + (0)(1)\cdot f(0,1) + (1)(0)\cdot f(1,0)] = 0$$

$$+ (1)(1)\,f(1,1) = f(1,1) = 0.3$$

Q3:

$$Corr(x,y) \neq \frac{Cov(x,y)}{\sigma_x\,\sigma_y}$$

$$Cov(x,y) = 3 \cdot 2 = 6$$

$$Cov(2x-3, 1-3y) = (2)(-3)\,Cov(x,y) = -36$$

Q4: for indep. r.vs $E(XY) = (EX)(EY)$.

$$EXY = 3^2 = 9.$$

Q5: $\frac{P(A \cap B)}{P(A)} = \frac{P(A \cap B)}{P(B)}$

$$\because P(A) = P(B) = 0.6$$

Question 7: For a random variable X with CDF $F(x) = 1 - e^{-2x}, x > 0$, find the pdf $f(x)$ and $P(X = 0.1)$ **(5 marks)**

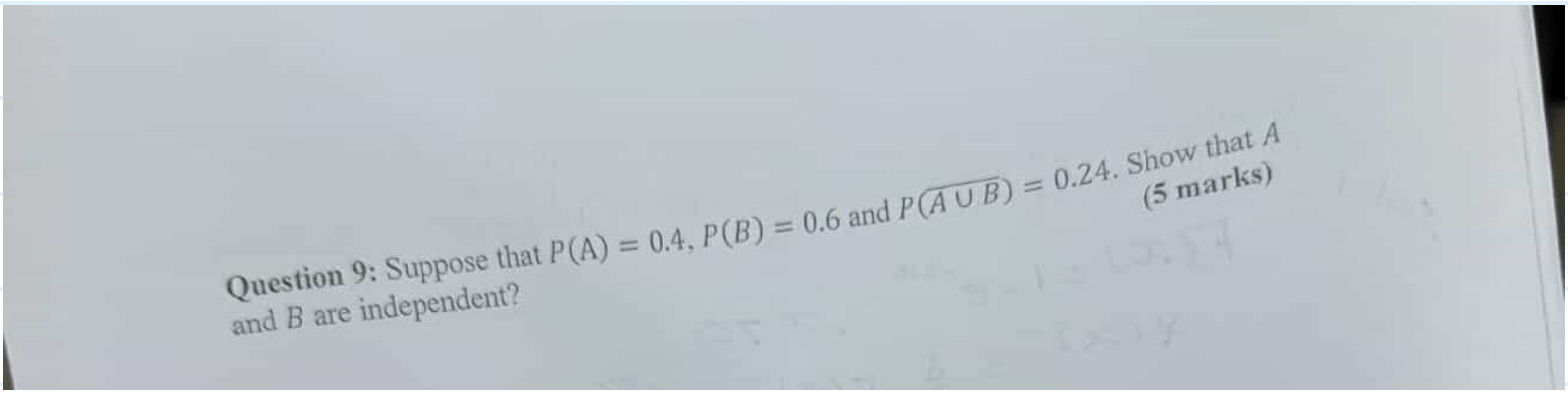
- $f(x) = \frac{dF(x)}{dx} = 2e^{-x}$ $\therefore$ $f(x) = \begin{cases} 2e^{-x}, & x > 0 \\ 0, & o.w \end{cases}$

- x is a continuous r.v, therefore:

$$P(X = 0.1) = 0.$$

Question 8: Consider the joint pdf $f(x,y) = 1$ when $(0 < x < 1,\ 0 < y < 1)$ and zero otherwise. Find $P(-1 < X < 1/2$ *and* $-1 < Y < 1/2)$. **(5 marks)**

$$\int_{-1}^{1/2}\int_{-1}^{1/2} 1\, dx\, d = \int_{-1}^{1/2} dx \cdot \int_{-1}^{1/2} dy = \int_{0}^{1/2} dx \cdot \int_{0}^{1/2} dy = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$



Question 9: Suppose that $P(A) = 0.4$, $P(B) = 0.6$ and $P(\overline{A \cup B}) = 0.24$. Show that A and B are independent? **(5 marks)**

by demorgin's Law:-

$$P(\overline{A \cup B}) = P(\bar{A} \cap \bar{B}) = 0.24$$

$$P(\bar{A}) = 1 - P(A) = 1 - 0.4 = 0.6$$

$$P(\bar{B}) = 0.4$$

iff A, B are indep. then

$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B})$$

$$0.24 \stackrel{?}{=} (0.4)(0.6) \quad \checkmark$$

∴ They are independent.